
Rock de Vocht

Senior Software Architect | Wellington, New Zealand | peter@peter.co.nz | 021 021 52225

Professional Summary

Innovative **Technical Leader and Architect** with a Master of Science in Computer Science and over 30 years of experience. Expert in designing **high-performance big-data retrieval systems**, complex **cloud infrastructure**, and **AI-driven architectures**. Proven track record of defining technical road-maps, managing the full SDLC, and mentoring engineering teams to deliver scalable enterprise solutions.

Core Competencies

- **Architectural Design:** Distributed Systems, Micro-services (Kubernetes/Docker), Semantic Search Engines.
 - **AI & NLP Integration:** LLM Context-Engineering, Neural Networks, Generative AI, and Machine Learning.
 - **Full-Stack Ecosystems:** JVM (Kotlin/Java), C#, Python, and React JS / Typescript.
 - **Cloud & Infrastructure:** Azure, Google Cloud, and Linux environment management.
 - **Leadership:** Team Leading, Mentoring, Solution Architecture, and IP Creation.
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Professional Experience

Chief Scientific Officer & Co-Founder | SimSage UK Ltd.

Jan 2020 – Present

- **System Architecture:** Engineered a novel, Graph-based Enterprise Semantic Search Engine utilizing advanced NLP for LLM context-engineering.
- **Integration Strategy:** Designed and maintained a massive code base (250k+ lines) featuring 30+ enterprise connectors including SharePoint and Google Drive.
- **AI Orchestration:** Architected production AI workflows integrated with Gemini, OpenAI, and CoPilot.
- **Infrastructure Optimization:** Leveraged Kubernetes, Docker, and Ubuntu Server to ensure scalable and resilient deployment environments.

Developer Contractor | MBIE (NZ Immigration)

May 2018 – Dec 2019

- **Solution Design:** Developed C# solutions on Azure (.Net framework) within Linux environments.
- **Data Architecture:** Managed the NLP processing of 170M unstructured text records for signal analysis and statistically significant data extraction.
- **Quality Engineering:** Implemented Machine Learning models for automated data quality and risk analysis.

NLP Specialist | Booktrack.com

Mar 2016 – May 2018

- **Strategic Advisory:** Advised the CTO on development strategies involving R&D in neural networks and machine learning.
- **System Design:** Designed Automatic Speech Recognition (ASR) systems to handle various accents and convert audio to text.
- **Cloud Orchestration:** Utilized Kubernetes and Docker across Google Cloud and Azure to support core IP development.

Chief Scientist & Co-Founder | SYL Search

May 2008 – Jan 2015

- **Product Vision:** Defined the technical road-map and led the development team through the full SDLC and patent-pending IP creation.
- **Government Solutions:** Created a semantic search engine adopted by major NZ Government departments, including MOT, NZTA, and MPI.

Senior Architect (Microsoft Team) | Datacom

Jan 2004 – May 2008

- **Software Architecture:** Served as a key member of the internal Architecture Solution Group, defining high-level technical strategies.
- **Team Leadership:** Acted as Team Lead and mentor, facilitating internal education sessions and providing technical pre-sales assistance.

Education & Certifications

- **Master of Science (Computer Science)** | University of Auckland
 - *Research:* Relationship between grammar, semantics, and symbolic logic.
- **Bachelor of Science (Computer Science) (Honours)** | University of Auckland
- **Machine Learning Certification** | Stanford University (via Coursera)
 - Updated knowledge on deep-learning neural networks and AI advancements.

Technical Skills & Languages

- **Languages:** Dutch, English, French, German.
 - **Code:** Kotlin, C#, Kotlin, Java, C, C++, Python, React.JS, TypeScript.
 - **DevOps/Cloud:** Azure, Kubernetes, Docker, Google Cloud, Ubuntu Server.
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